Enhancing Access and Reproducibility of AML Flow Cytometry Data Analysis through Machine Learning Based Automated Cross-Panel Classification at Sample and Cell Level

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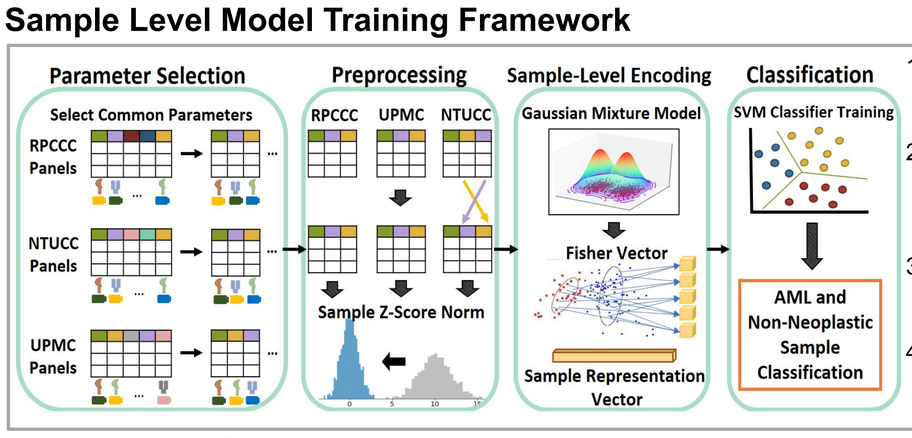
Introduction

- There is considerable variation among cytometry reagents, instruments and analysis approaches used for hematological disease diagnosis and monitoring.
- We previously demonstrated that our approach is adaptable across different instruments for sample level classification model development [1,2].
- In this study, we combined **three different panels: ClearLLab10C, Euroflow AML/MDS and a UPMC diagnostic panel** to demonstrate the feasibility of developing **sample and cell level classification models thats instrument and reagent agnostic**.

Experiment: Cross-Site Sample Level

Sample Level Model Training Framework



1. Overlapped parameters selected to conduct sample level z-score normalization.
2. Gaussian mixture model (GMM) captures cellular distribution, followed by Fisher vectorization for embedding into a representation vector.
3. Representations input into support vector machine (SVM) classifier for AML vs non-neoplastic classification.
4. Dataset divided into training and testing sets and used for 5-fold cross-validation.

Sample Level Classification Model Performance

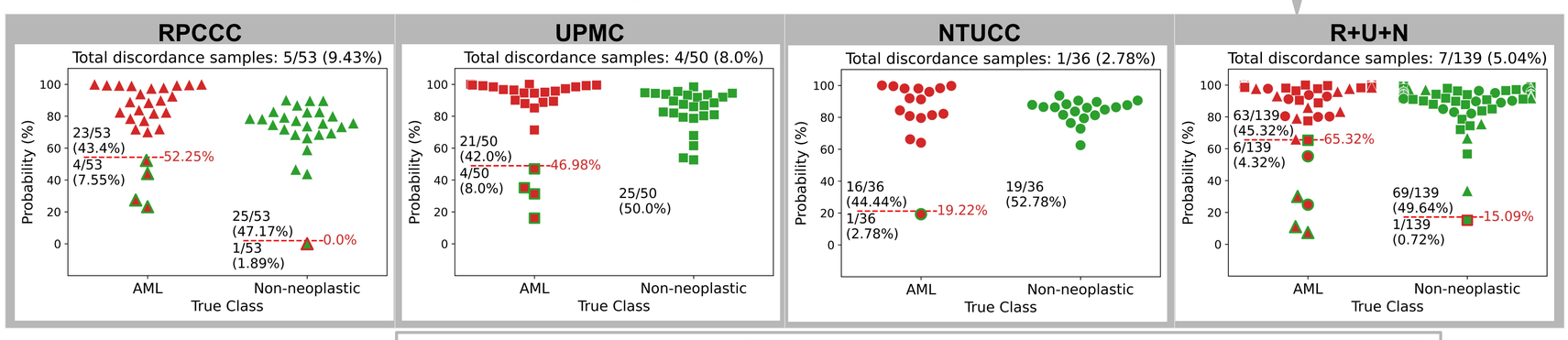
Parameters	33	37	7	7 (overlapped parameters* between RPCCC, UPMC and NTUCC)						
Training Dataset	RPCCC (R)	UPMC (U)	NTUCC (N)	RPCCC (R)	UPMC (U)	NTUCC (N)	R+U+N	R+U+N		
Testing Dataset								RPCCC	UPMC	NTUCC
ACC	100.00%	95.65%	97.22%	90.57%	92.00%	97.22%	94.34%	96.00%	94.44%	94.96%
AUC	100.00%	95.81%	98.76%	92.88%	97.12%	98.76%	98.15%	99.68%	100.00%	98.59%
Sensitivity	100.00%	96.00%	100.00%	96.15%	100.00%	100.00%	100.00%	96.00%	100.00%	98.57%
Specificity	100.00%	95.24%	94.12%	85.19%	84.00%	94.12%	88.89%	96.00%	88.24%	91.30%

Overlapped parameters*: FSC-A, FSC-H, SSC-A,CD34,CD45,CD117,HLA-DR
ACC: accuracy; AUC: area under the ROC curve

Sample Level Dataset

Dataset	RPCCC (R)	UPMC (U)	NTUCC (N)	R+U+N
FC record	53	50	36	139
Flow Panel	ClearLLab10C	UPMC Diagnostic	Euroflow AML/MDS	
Instrument				
FASCanto	0	50	0	50
Navios EX	53	0	0	53
FACSLyric	0	0	36	36
Specimen				
PB	5	0	0	5
BM	48	50	36	134
Diagnosis				
AML	27	25	17	69
non-neoplastic	26	25	19	70

Sample Level Prediction Probability Ranking



Abbreviations:
FC: Flow Cytometry; PB: Peripheral Blood; BM: Bone Marrow; AML: Acute Myeloid Leukemia; RPCCC: Roswell Park Comprehensive Cancer Center; NTUCC: National Taiwan University Cancer Center

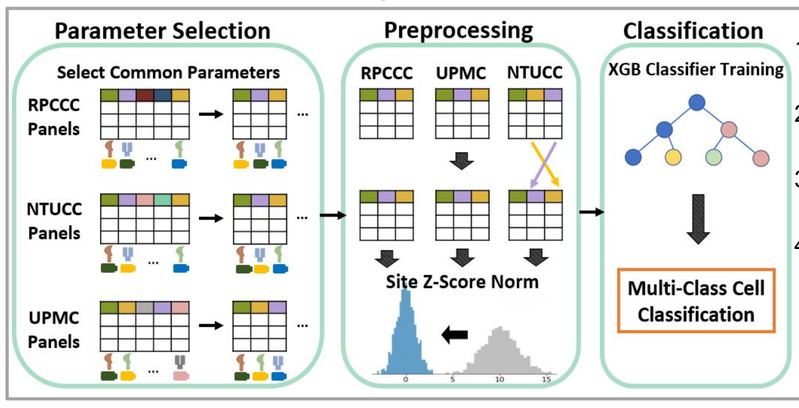
Legend

RPCCC	UPMC	NTUCC
△	□	○

		Predict Label	
		AML	Non-neoplastic
Diagnosis	AML	■	■
	Non-neoplastic	■	■

Experiment: Cross-Site Cell Level

Cell Level Model Training Framework



1. Overlapped parameters selected to conduct site z-score normalization.
2. Adjust the quantities of each dataset and each class to be equal dimension prior to training.
3. Utilize the XGBoost classifier for multi-class classification.
4. Implemented 5-fold cross-validation, dividing dataset into training and testing sets based on samples.

Cell Level Classification Model Performance

Parameters	14	14	11	8 (overlapped parameters* between RPCCC, UPMC and NTUCC)						
Training Dataset	RPCCC (R)	UPMC (U)	NTUCC (N)	RPCCC (R)	UPMC (U)	NTUCC (N)	R+U+N	R+U+N		
Testing Dataset								RPCCC	UPMC	NTUCC
AUC	99.70%	98.50%	97.60%	99.50%	98.20%	96.40%	97.10%	99.40%	98.00%	96.40%
ACC	95.50%	92.70%	92.20%	94.20%	90.60%	88.60%	88.30%	93.50%	89.70%	87.50%
UAR	96.70%	94.30%	95.00%	95.90%	91.90%	94.50%	93.90%	95.90%	91.80%	94.10%
Weighted F1	95.70%	92.80%	92.30%	94.60%	91.00%	89.80%	89.10%	93.90%	90.20%	88.70%

overlapped parameters*: FSC-A, FSC-H, SSC-A,CD11b, CD13, CD16, CD45, HLA-DR
AUC: area under the ROC curve; ACC: accuracy; UAR: Unweighted Average Recall

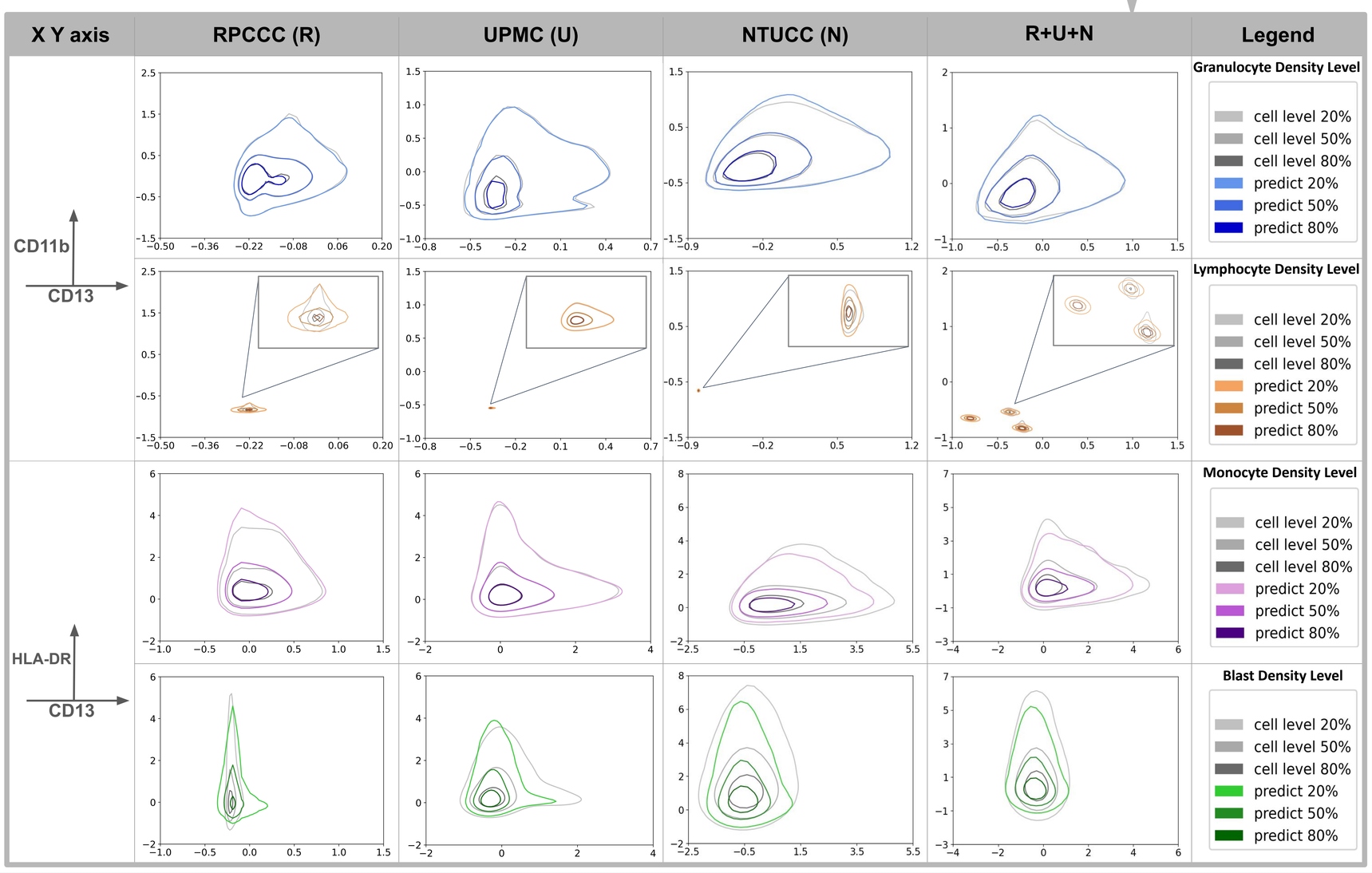
Cell Level Dataset

- All cell-level classification samples are from BM specimens, diagnosed as non-neoplastic conditions.
- ClearLLab10C M1 tube and UPMC panel tube 1 and Euroflow AML/MDS panel tube 1 were used.

Dataset	RPCCC (R)	UPMC (U)	NTUCC (N)	R+U+N
Sample size	21	22	10	53
Flow Panel	ClearLLab10C	UPMC Diagnostic	Euroflow AML/MDS	
Cell Label				
Total class	1,480,519	659,952	9,999,993	12,140,464
Granulocyte	569,711	377,267	4,133,242	5,080,220
Lymphocyte	179,841	96,706	18,191	294,738
Monocyte	29,680	14,323	84,365	128,368
Blast	14,094	5,749	19,577	39,420
Others	687,193	165,907	5,744,618	6,597,718

Density Contour Comparison between Cell Label and Prediction

Training and testing dataset using common parameters. (FSC-A, FSC-H, SSC-A,CD11b, CD13, CD16, CD45, HLA-DR)
The X-axis and Y-axis values have both been normalized via site z-score transformation.



Conclusions

Summary

- Our approaches allow flexibility using data from three different reagent panels and instrument manufacturers.
- The GMM-SVM approaches successfully developed both sample in distinguishing between AML and non-neoplastic samples.
- Our approach successfully develop cell type classification models to distinguish mature leukocytes from immature blasts.

Future Work

- Investigate further in discordance cases to improve data normalization and classification model performance.
- Expand the training dataset to cover wider spectrum of reagent panels and instrument models.

Supplementary Data

References

1. Wang Y-F, Li J-L, Lee C-C, Wallace PK, Ko B-S: Flow Cytometry Protocols. 2024. pp. 353-67.
2. Ko, B.S., et al. EBioMedicine 37, 91-100 (2018).
3. Li, J.L., et al. Annu Int Conf IEEE Eng Med Biol Soc 2019, 1733-1736 (2019).
4. Monaghan, S.A., et al. Am J Clin Pathol 157, 546-553 (2022).

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